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SUPPLEMENTARY INFORMATION
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Multinucleotide mutations cause false inferences of lineage-specific positive selection

Aarti Venkat[1], Matthew W. Hahn[2,3] and Joseph W. Thornton[1,4]*

[1]Department of Human Genetics, University of Chicago, Chicago, IL, USA. [2]Department of Biology, Indiana University, Bloomington, IN, USA. [3]Department of Computer Science, Indiana University, Bloomington, IN, USA. [4]Department of Ecology and Evolution, University of Chicago, Chicago, IL, USA.
*e-mail: joet1@uchicago.edu

SUPPLEMENTARY TABLES AND FIGURES

for Venkat A, Hahn MW, Thornton JW. "Multinucleotide mutations cause false positive inferences in lineage-specific tests of positive selection."

Supplementary Table 1. Paths between codon pairs

Average steps on path (nonsyn, syn)	Number of pairs	Proportion of all pairs (%)
0, 2	8	0.6
0.5, 1.5	0	0.0
1, 1	588	40.5
1.5, 0.5	308	21.2
2, 0	548	37.7

For every codon pair separated by 2 nucleotide differences, the universal genetic code was parsed to tabulate the mean number of nonsynonymous and synonymous steps (nonsyn, syn) on the two direct paths between them. Paths with stop codons were excluded.

Supplementary Table 2. Number of gene alignments passing each filtering step

	Humans			Flies		
	Empirical	Simulated	Control	Empirical	Simulated	Control
All genes	16,541	6,868	6,868	8,564	8,564	8564
Genes with complete species coverage	6,868	6,868	6,868	8,564	8,564	8564
Tests significant at P<0.05 (% of tests)	82 (1.1%)	99 (1.4%)	32 (0.5%)	3,938 (7.6%)	4,444 (8.6%)	582 (1.1%)
Tests significant after correction (FDR <0.2)	0	0	0	2,147	1,755	4
BST-significant genes (tests) with unambiguous ancestral codon reconstructions	30 (30)*	-	-	443 (458)	-	-

Filtering steps are described in Methods. Empirical data are all genome-wide coding alignments from the studies by Kosiol *et al*., and Larracuente *et al*.[12,44] Simulated data are alignments simulated under the BS+MNM null model using parameters derived from the empirical data. Control data are alignments simulated as above but with MNM rate parameter $\delta = 0$. *, in humans, the empirical BST-significant set includes genes that pass the filter for ancestral reconstruction of CMDs but not for FDR adjustment; in flies, both criteria are met. The total number of tests on the 6 fly lineages is 51,384.

Supplementary Table 3. The BS+MNM null model fits many empirical genes better than the BST null model.

Lineage	Genes tested	Genes significant (%) Empirical genes	Control simulations
human	6868	1545 (22%)	162 (2%)
D. melanogaster	8564	5080 (60%)	203 (2%)
D. simulans	8564	5089 (59%)	253 (3%)
D. sechellia	8564	4994 (58%)	236 (3%)
D. yakuba	8564	5089 (59%)	251 (3%)
D. erecta	8564	4957 (58%)	210 (2%)
D. ananasae	8564	4297 (51%)	217 (2%)

The table shows the number of empirical genes tested on each lineage and the number and proportion that are significantly better fit by the BS+MNM null model than the BST null model, using a likelihood ratio test ($P<0.05$, df=1). For the control simulations, the false positive inference rate was assessed by simulating sequences using the parameters of the BST null model inferred from each empirical gene, then analyzing these sequences with the same likelihood ratio test. After adjustment for multiple testing (FDR<0.2), only one simulated gene on the *D. melanogaster* branch and 2 genes on the *D. erecta* branch were significant; for all other lineages, zero genes led to positive results.

Supplementary Table 4. Proportion of tests that lost and retained significance after the BS+MNM test was applied to BST-significant genes.

Species	Lost significance	Retained significance	Retained significance (triple)
Human	28/30 (94%)	2/30 (6%)	0/30 (0%)
Flies	174/458 (38%)	213/458 (47%)	71/458 (16%)

BST-significant tests in the empirical datasets were reanalyzed using the BS+MNM test. The number and fraction of tests that lost or retained statistical significance ($P<0.05$) are shown. The third column shows the proportion of tests that retained significance for which the gene contains one or more codons with 3 differences, which require multiple substitutions in both the BST and BS+MNM models.

Supplementary Table 5. Observed frequency of tandem substitutions on the human and *D. melanogaster* lineages in both empirical and simulated datasets.

	Empirical	Simulated
Humans	295 / 23,335 (1.26%)	326 / 20,040 (1.63%)
D. melanogaster	3,895 / 249,344 (1.56%)	9,069 / 278,978 (3.25%)

The number of tandem substitutions at adjacent nucleotide sites is shown as a fraction of all substitutions that occurred on the terminal lineages leading to human and *D. melanogaster*. For the empirical data, we counted the number of sequence differences between the extant sequence and the phylogenetically reconstructed sequence of the human-chimp ancestor or *D. melanogaster*/*D. simulans* ancestor. For the simulated data, sequences were simulated under the BS+MNM null model and the parameters estimated from each empirical gene; substitutions were counted as the number of differences between the terminal and ancestral sequences at the same nodes as in the empirical analysis. Tandem differences at adjacent sites were counted as a single substitution event.

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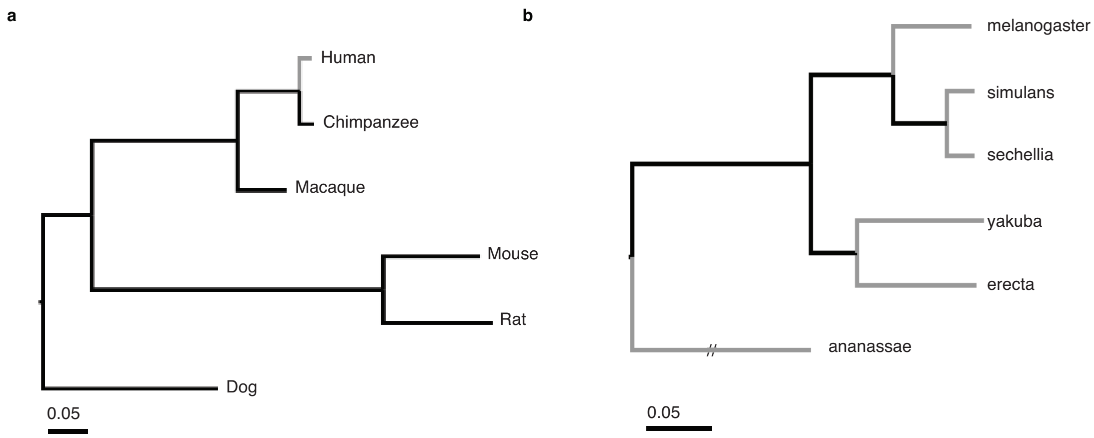
Supplementary Table 6. Filtering based on ancestral state reconstruction does not affect CMD enrichment.

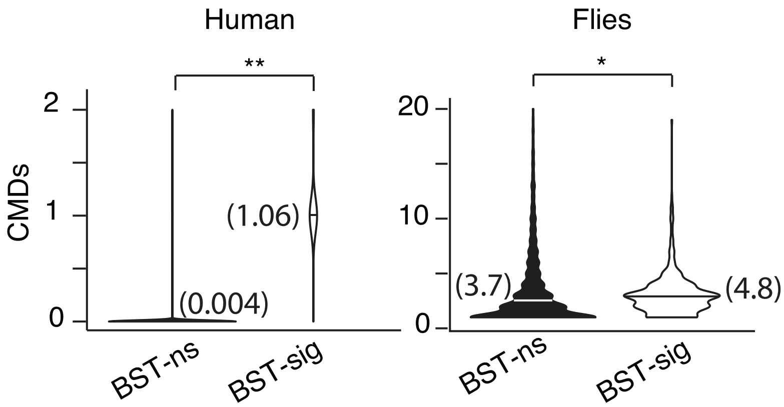
	CMDs	non-CMDs	CMDs / non-CMDs
BST+ (filtered)	32	19,563	0.00163
BST+ (non-filtered)	35	20,923	0.00167

The table shows the number and ratio of CMDs and non-CMDs in the original set of 30 filtered BST-significant genes on the human lineage – from which genes with ambiguously reconstructed CMDs were filtered out. These quantities are also shown for the larger set of 33 BST-significant genes when the filter is not applied.

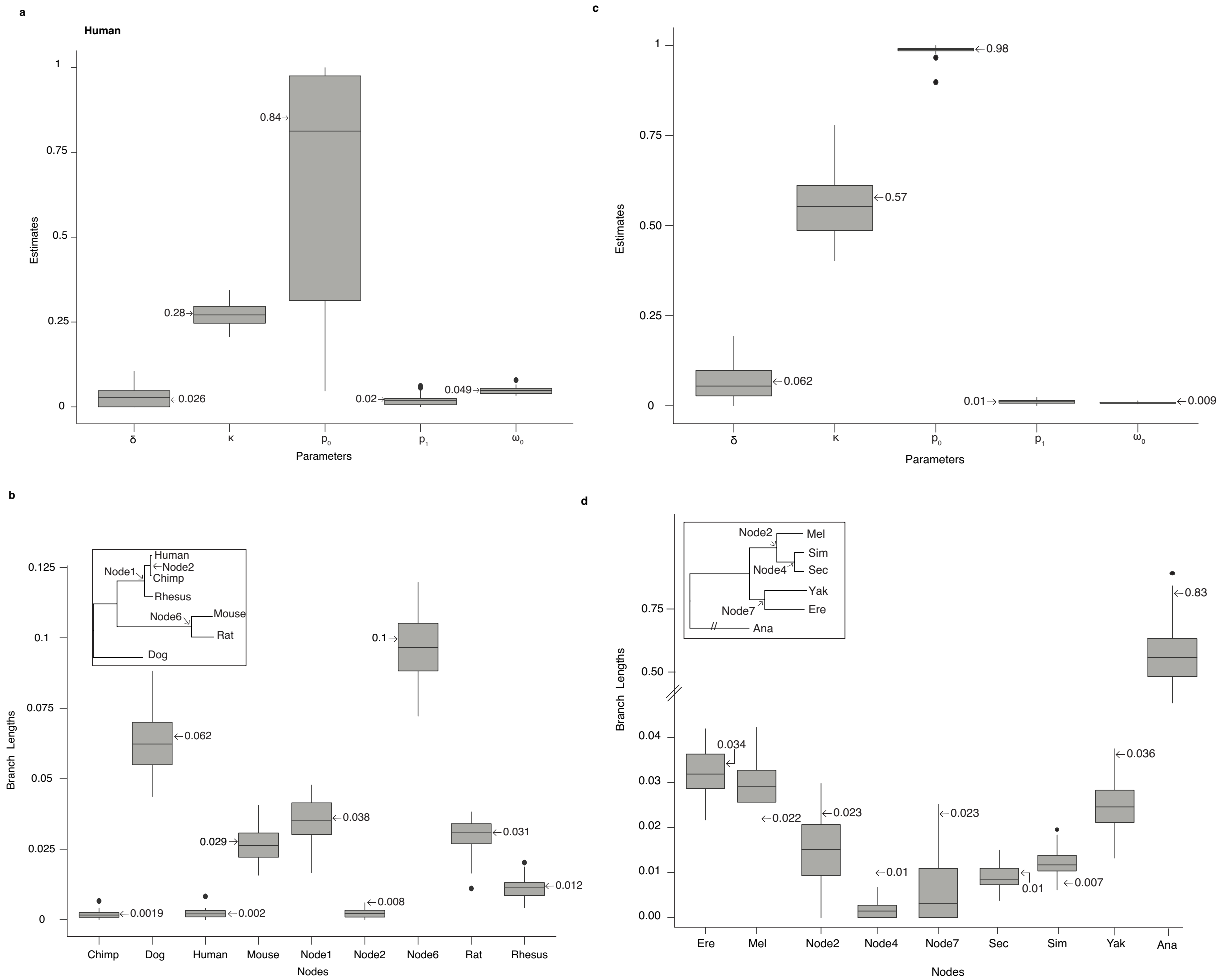


Supplementary Figure S1. Phylogenies of mammalian and *Drosophila* species used in this analysis.

The BST was used to identify genes under positive selection on each of the lineages colored in grey. Branch lengths are proportional to the median length across all genes analyzed. Scale bar, expected nucleotide substitutions per codon; the hatched branch, shortened for display, has length 0.62.

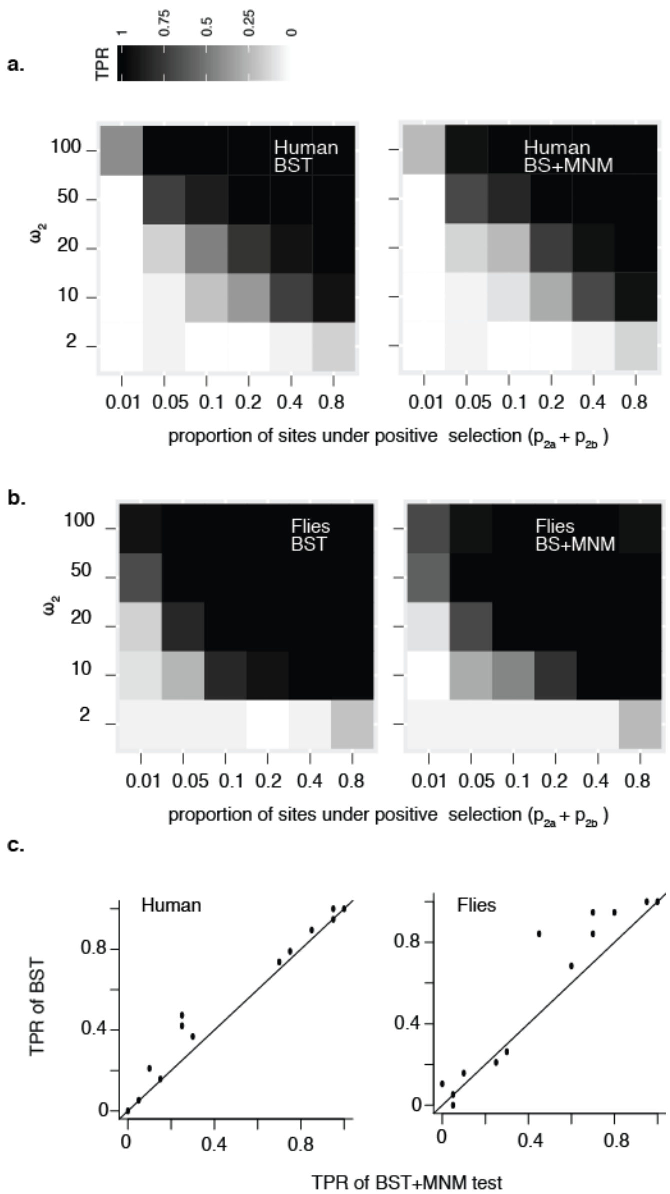


Supplementary Figure S2. Distribution of the number of CMDs per gene in BST-significant (BST-sig) and BST-nonsignificant (BST-ns) genes. Horizontal line in each violin plot indicates the median. *, P=4e-10; **, P<2e-16, Mann Whitney U test.



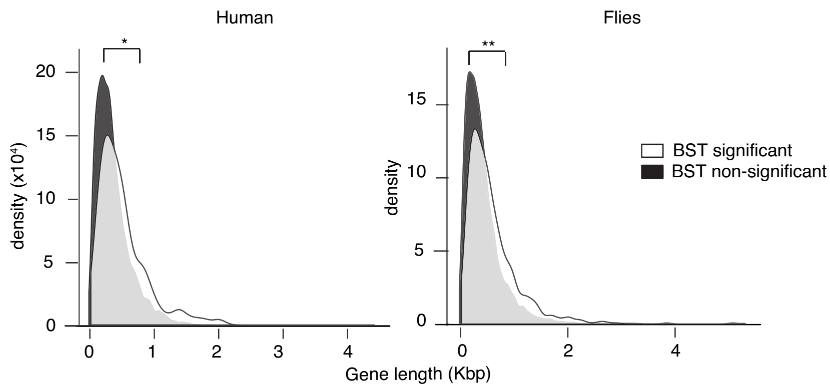
Supplementary Figure S3. Validation of parameter estimation in the BS+MNM model.

50 replicate alignments were simulated under the BS+MNM null model with genome-median parameters, including gene length, for the mammalian (**a,b**) and fly **(c,d)** datasets; the BS+MNM null model was then used to estimate the parameters from each replicate. **(a,c)** Box plots show the distribution of ML estimates across replicate alignments for model parameters: δ, the MNM rate multiplier; κ, the transversion:transition rate multiplier; p_0, mixture weight for submodel 0; ω_0, nonsynonymous/synonymous rate multiplier for submodel 0; p_1, mixture weight for submodel 1. **(b,d)** Distribution of ML estimates across alignments for branch lengths, labeled by the node at the end of each branch on the phylogeny shown in the inset. Arrows indicate the generating parameters used in the simulation.



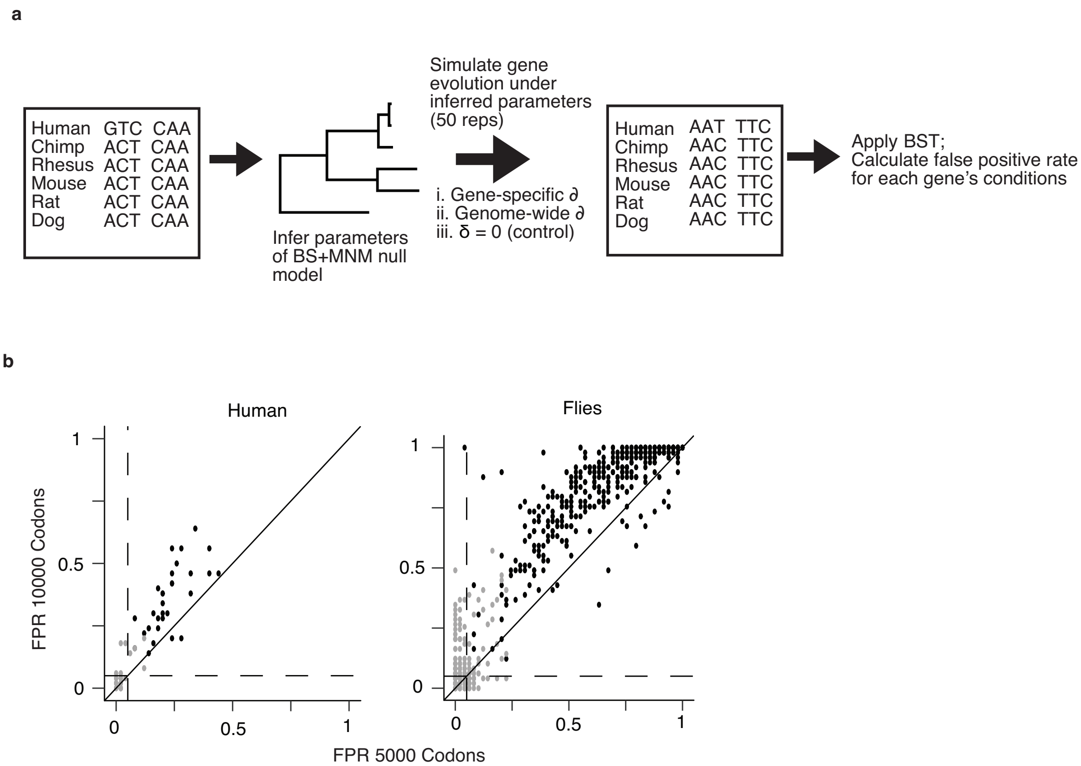
Supplementary Figure S4. Analysis of power to detect positive selection by the BST and BS+MNM test under empirically derived conditions.

Replicate sequence alignments of genome-median length were simulated under the BST model with positive selection on the human **(a)** or *D. simulans* **(b)** branches; the proportion of sites under positive selection on the branch (p_2) and the strength of positive selection (ω_2) were varied, and all other conditions, including branch lengths, were set to their genome-median values. These alignments were then tested for positive selection using the BST (left) or BS+MNM test (right). Each cell in the grid shows the true positive inference rate (TPR at $P<0.05$), shaded according to the heat map, for 20 replicates under that set of conditions. **(c)** Scatter plot comparing power of the BS+MNM test and BST; each dot represents the TPR of the two methods under one of the conditions in panels (a) and (b).



Supplementary Figure S5. Longer genes are more likely to yield false positive BST results.

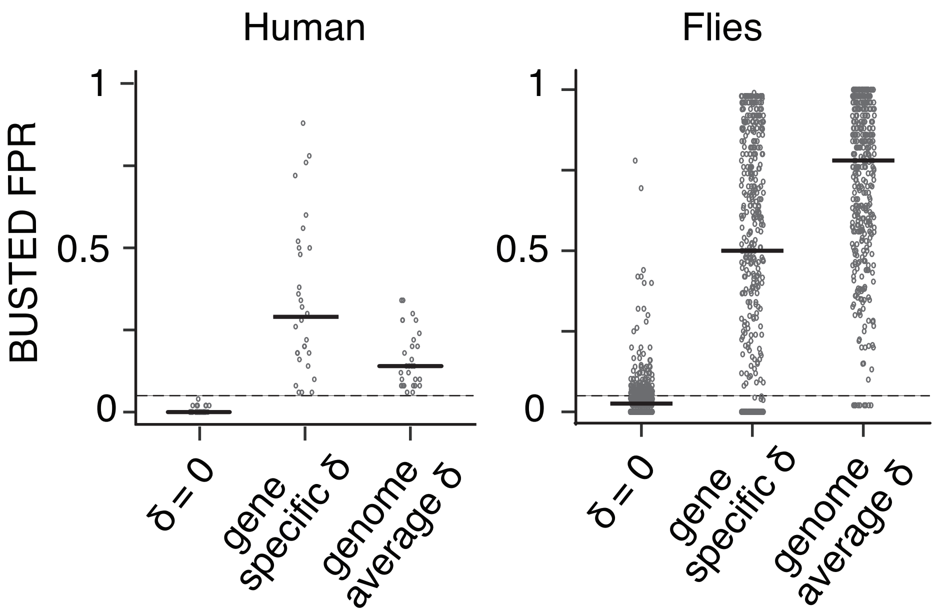
For each empirical gene in the mammalian and fly datasets, the parameters of the BS+MNM null model were estimated by maximum likelihood. We then simulated sequence evolution under each gene's inferred null parameters and empirical length and used the classic BST on the simulated alignments to test for positive selection on the human and terminal fly lineages. The distribution of the lengths of genes yielding a BST-significant test ($P<0.05$) or a BST-nonsignificant test is shown. Median lengths in BST-significant and non-significant genes, respectively, were 422 and 343 bp in humans; in flies, 484 and 391 bp. Differences between distributions were evaluated using Mann-Whitney U test. *, P=5e-3; **, P=2e-23.



Supplementary Figure S6. MNMs bias the classic BST.

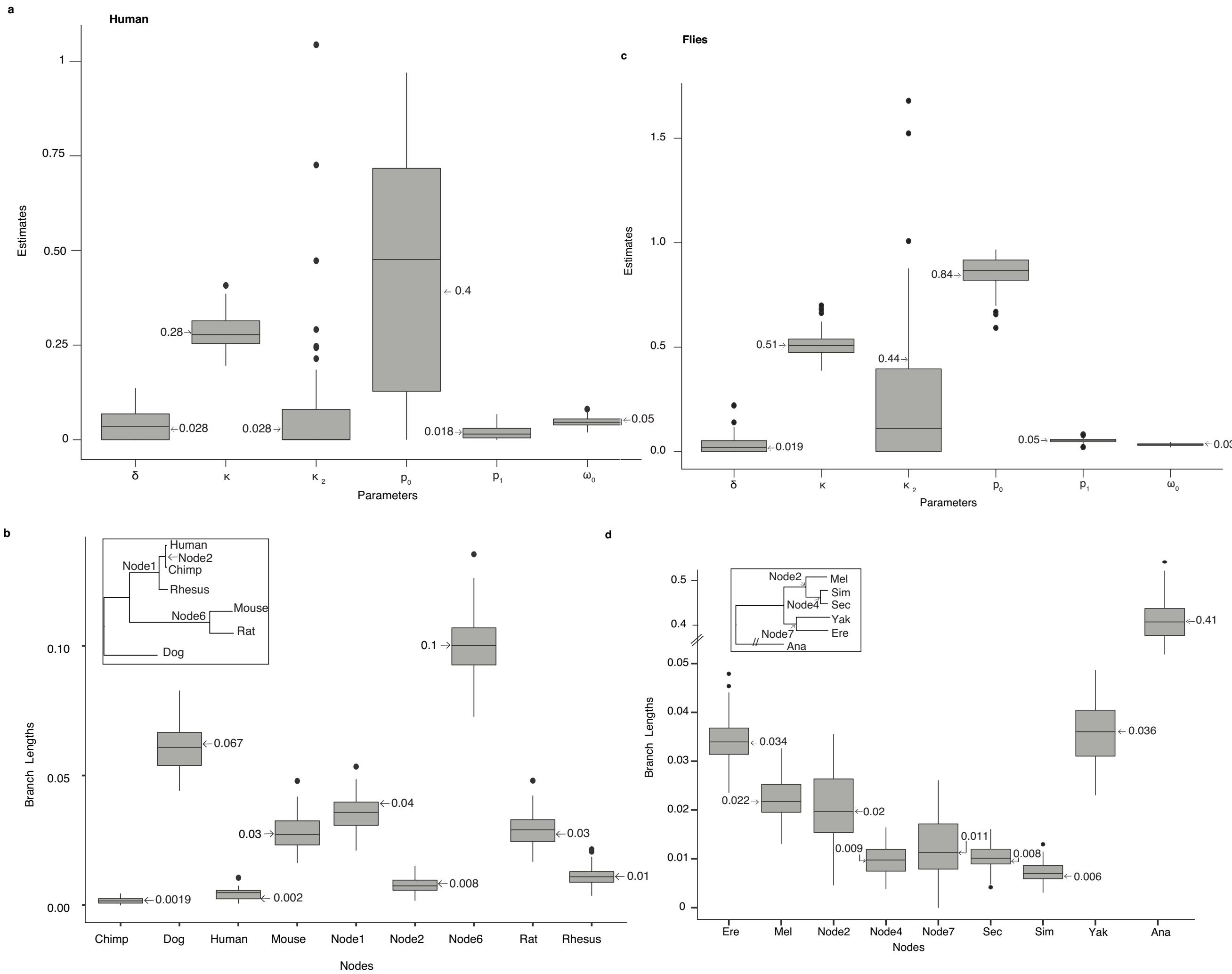
(a) Scheme to simulate genes with MNMs without positive selection. For each BST-significant empirical gene, the parameters of the BS+MNM null model were estimated. Using each set of parameters, 50 replicate alignments were simulated, with δ (the relative rate of MNM substitution) assigned to its gene-specific value, its median across all genes in the dataset (genome-wide average), or to zero. The classic BST was applied to simulated data, and the false positive rate (FPR) for each set of generating parameters was calculated as the fraction of replicates with a positive result ($P<0.05$).

(b) Systematic bias in the BST using the genome-average MNM rate. 50 replicate alignments 5,000 or 10,000 codons long were simulated under the BS+MNM null model using gene-specific parameters inferred as in (a). Each black point represents FPRs for sequences 5,000 and 10,000 codons long simulated under one empirical gene's parameters and the genome-wide average δ. Gray points show FPRs for control simulations with $\delta = 0$. Dashed lines, FPR of 0.05. Diagonal line has a slope of 1.



Supplementary Figure S7. MNMs cause bias in BUSTED, a newer test of positive selection.

For every BST-significant gene in each dataset, 50 replicate alignments 5,000 codons long were simulated using the BS+MNM null model and parameter values estimated from the empirical sequences. These alignments were then analyzed for a signature of positive selection ($P<0.05$) using BUSTED. δ was assigned to its gene-specific estimate, to its average across all genes in each dataset, or to zero. FPR is the proportion of replicate alignments for each gene with $P<0.05$. Each dot represents the FPR for one gene; black bars are the median across genes.



Supplementary Figure S8. Validation of parameter estimates by BS+MNM+κ_2 model.

50 replicate alignments were simulated under the BS+MNM+ κ_2 null model using genome-wide median parameters. Parameters were then re-estimated given each alignment using the same model. Box plots show the distribution across replicate alignments of ML estimates of model parameters (**a,c**) and branch length (**b,d**) in humans (**a,b**) and in flies (**c, d**). For detailed explanation of model parameters, see **Fig. S3**. κ_2 is the transversion:transition rate ratio for CMDs. Arrows show the genome-wide median used to simulate sequences. The median value of $\boldsymbol{\kappa_2}$ is less than the median $\boldsymbol{\kappa_1}$, whereas the mean $\boldsymbol{\kappa_2}$ is greater than the mean $\boldsymbol{\kappa_1}$. This difference is likely caused by the rarity of CMDs, especially those with multiple transversions in each codon, which leads to large variance and uncertainty in the estimate of $\boldsymbol{\kappa_2}$ in many individual genes.